BOOK RECOMMENDATION SYSTEM

What book you should read next?

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INTRODUCTION

Have you ever wondered how Spotify or YouTube seems to understand your taste so well? These systems recommend content with surprising accuracy, and we were curious about the methods behind them. As book lovers ourselves, we wanted to explore how similar recommendation techniques could be applied to books.

This project aims to build a Book Recommendation System that suggests what to read next based on your input reading by using collaborative filtering

METHODOLOGY

We implemented **item-item collaborative filtering**, focusing on the similarities between books. The process includes:

- Data Pre-process: Replaced rows zero ratings with NAN values.
- Filtering: Only included users with more than 10 ratings and books with more than 4 ratings
- Merging: Combined the ratings and books metadata to enrich our dataset.
- Matrix Construction: Created a pivot table of users vs. books and filled NaN values with zeros.
- Similarity Calculation: Used cosine similarity to find similar books based on user rating patterns.
- Recommendation: For a given book, return the top-N most similar books.

COLLABORATIVE FILTERING

1. MATRIX CONSTRUCTION

- Created a user-item rating matrix R, where:
  - Rows = book titles
  - Columns = users ID
  - Entries r_ij= rating given by user j to book i.
- Contained NaN values, indicating the absence of a rating for that book

Book Nam User ID	27625	27626	27627	27628	27629	27630	$\bar{i}$
Classical	10	?	?	1	?	1	4
Clara Call	7	?	2	?	4	7	5
Decision i	1	2	?	?	4	5	3
PLEADING	?	4	?	6	2	?	4
Jane Doe	?	3	6	9	?	?	6

2. SIMILARITY MATRIX

a. Normalize (mean-centering):

- For each book, subtract the average rating of that book from each rating:

$$\hat{r}_{ij} = r_{ij} - \bar{r}_i$$

- This removes bias from universally high/low-rated books.

Book Nar User ID	27625	27626	27627	27628	27629	27630
Classical	6	0	0	-3	0	-3
Clara Ca	2	0	-3	0	-1	2
Decision	-2	-1	0	0	1	2
PLEADING	0	0	0	2	-2	0
Jane Doe	0	-3	0	3	0	0

b. Cosine Similarity:

- Measure the consine of the angle between two normalised vectors.
- Range: [-1;1]

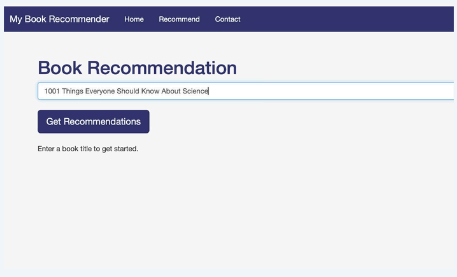
$$\text{cosine_similarity}(\mathbf{i}_1, \mathbf{i}_2) = \cos(\mathbf{i}_1, \mathbf{i}_2) = \frac{\mathbf{i}_1^T \mathbf{i}_2}{\|\mathbf{i}_1\|_2 \|\mathbf{i}_2\|_2}$$

With +1: most similar, 0: no similarity, -1: least similar

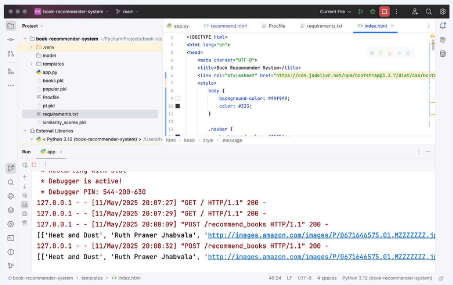
Book	Classical	Clara Ca	Decision	PLEADING	Jane Do
Classical	1	0.400	-0.431	-0.327	-0.707
Clara Cal	0.400	1	-0.588	-0.267	-0.298
Decision i	-0.431	-0.588	1	0.000	-0.134
PLEADING	-0.327	-0.267	0.000	1	0.949
Jane Doe	-0.707	-0.298	-0.134	0.949	1

RESULT AND DEMO

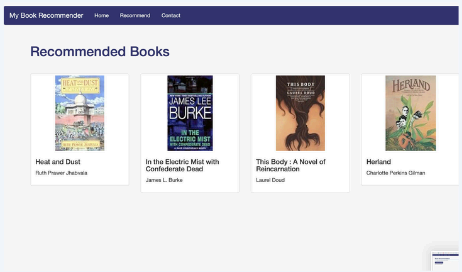
1. User inputs book title



2. System processes input and computes recommendations



3. Recommendations result displayed on the website



FUTURE IMPLICATIONS

- Extend the system to predict personalized ratings instead of only suggesting based on a single book input.
- Apply the following item-based collaborative filtering formula to estimate unknown ratings:

$$\hat{r}_{ui} = \frac{\sum_{j \in S(i)} \text{sim}(i,j) \cdot r_{uj}}{\sum_{j \in S(i)} |\text{sim}(i,j)|}$$

- This approach will allow the system to recommend books even if the user hasn't rated them, by leveraging rating patterns of similar items.
- Potential future work includes user profiling, session-based recommendations, and integrating deep learning for better personalization.

CONCLUSION

- Applied collaborative filtering to recommend books based on shared user rating patterns.
- Used matrix operations and cosine similarity to measure book-to-book similarity.
- System suggests relevant books from a single input title — no user profiles required.
- Demonstrates the practical role of linear algebra in real-world recommendation systems.

REFERENCES

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